

Differential Splicing with Subisoform Covariance

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?
2. Does local splice-path usage differ across cell types within the same mice?

The primary analysis aggregates cells into *mouse-by-condition-by-cell-type pseudobulks*. An evaluated alternative retains each cell's EC counts and isoform baseline in the local likelihood but estimates one shared path effect for the pseudobulk. Both approaches keep biological replication at the mouse level; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length isoforms are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0 : \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed only by a nonredundant splice-path partition. It is an omnibus test of equal path usage across the included cell types, after absorbing mouse and condition into a mouse-specific fixed effect. Cell types must be represented in at least three mice, and the primary FDR family requires at least 30 pseudobulk observations per partition. These two hypothesis families are reported separately.

Workflow

The analysis proceeds in five steps:

1. Construct local splice blocks from the isoform annotation and collapse isoforms to paths within each block.
2. Aggregate paired oligo(dT) and random-hexamer EC counts, and hierarchical isoform estimates, within each mouse, condition, and cell type.
3. Refine path usage in each pseudobulk with the exact paired-primer EC likelihood. Optionally, retain separate cell likelihood terms while sharing one pseudobulk path perturbation.

4. Estimate path-logratio covariance from EC ambiguity and UMI sampling. Estimates with weak information or boundary path usage are retained in output but excluded from Gaussian testing.
5. Fit mouse-replicate GLS condition models and compositional cell-type models. Test condition within cell type and cell type within mouse as separate hypothesis families.

Two inferential covariance calculations are reported. The *conditional* calculation fixes the fitted isoform mixture within each path. The *profile* calculation allows all isoform directions to vary as nuisance parameters. They are sensitivity analyses for the same block-by-cell-type hypothesis, not independent biological hypotheses.

Notation and Dimensions

An *equivalence class* (EC) is a set of isoforms compatible with one UMI. An EC count is therefore observed, whereas its allocation among compatible isoforms is latent. A *splice path* is an ordered local exon chain between two constitutive anchors. A *subisoform* is the set of full-length isoforms taking the same path through one block. PSI means percent spliced in; for two paths it is one component of the path-composition vector. ILR means isometric log-ratio, an orthonormal coordinate system for a composition.

Let $g = g(b)$ denote the gene containing block b , and let $\mathcal{I}_g$ be the set of isoforms retained by the paired-primer model and assigned to gene g by the isoform-to-gene table. Define $T_g = |\mathcal{I}_g| \in \mathbb{N}$. This is a gene-level set, not the set of isoforms whose exons overlap the block, and it is therefore shared by every block of the gene. Let $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$ be the retained, annotated spliced isoforms represented in the block's GTF-derived path map. Isoforms in $\mathcal{I}_g \setminus \mathcal{I}_b^{\text{sp}}$, including modeled unspliced precursors, remain nuisance isoforms in the EC likelihood.

Let $n_b, S_b \in \mathbb{N}$ be the numbers of qualifying pseudobulks and represented spliced paths for block b , with $S_b \geq 2$. Then $d_b = S_b - 1 \in \mathbb{N}$ is its number of independent log-ratios. There are $P = 2$ primer protocols. Primer p has $E_{bp} \in \mathbb{N}$ ECs for gene $g(b)$. Pseudobulk i contains $H_{bi} \in \mathbb{N}$ cells. The principal symbols are:

symbol	dimension or domain	meaning
b	$\{1, \dots, \mathcal{B}\}$	splice-block index
i	$\{1, \dots, n_b\}$	pseudobulk index within block b
p	$\{1, \dots, P\}$	primer-protocol index
h	$\{1, \dots, H_{bi}\}$	cell index within pseudobulk i
t	$\{1, \dots, T_g\}$	isoform index within gene $g(b)$
s	$\{1, \dots, S_b\}$	splice-path index
$\mathbf{c}_{bip}$	$\mathbb{N}_0^{E_{bp}}$	pseudobulk EC counts
A_{bp}	$\mathbb{R}_+^{E_{bp} \times T_g}$	weighted EC-by-isoform design
$\boldsymbol{\theta}_{bi}$	$\mathbb{R}_+^{T_g}$	isoform abundance
C_b	$\{0, 1\}^{S_b \times T_g}$	isoform-to-path map
Q_b	$\mathbb{R}^{S_b \times d_b}$	Helmert ILR basis
$\boldsymbol{\delta}_{bi}$	$\mathbb{R}^{d_b}$	local path perturbation
$\boldsymbol{\psi}_{bi}$	$\Delta^{S_b - 1}$	path composition
$\mathbf{y}_{bi}$	$\mathbb{R}^{d_b}$	ILR response
V_{bi}	$\mathbb{S}_+^{d_b}$	ILR inferential covariance

Here $\mathcal{B} \in \mathbb{N}$ is the number of blocks, $\mathbb{N}_0 = \{0, 1, \dots\}$, $\Delta^{S_b-1} = \{\mathbf{v} \in \mathbb{R}_+^{S_b} : \mathbf{1}'\mathbf{v} = 1\}$, and $\mathbb{S}_+^{d_b}$ is the set of symmetric positive-semidefinite $d_b \times d_b$ matrices. Bold lower-case symbols are vectors, upper-case symbols are matrices, $\mathbf{1}_k \in \mathbb{R}^k$ is a vector of ones, $\mathbf{0}_k \in \mathbb{R}^k$ is a vector of zeros, $I_k \in \mathbb{R}^{k \times k}$ is the identity, $\odot$ is elementwise multiplication, and $(\cdot)^+$ is the Moore–Penrose pseudoinverse.

Helmert basis and ILR coordinates

The simplex has only $d_b = S_b - 1$ free directions because path proportions sum to one. The Helmert matrix Q_b is a fixed orthonormal basis for the sum-zero subspace:

$$Q_b' Q_b = I_{d_b}, \quad Q_b' \mathbf{1}_{S_b} = \mathbf{0}_{d_b}.$$

Using one-based row $r \in \{1, \dots, S_b\}$ and column $j \in \{1, \dots, d_b\}$, its entries are

$$(Q_b)_{rj} = \begin{cases} 1/\sqrt{j(j+1)}, & r \leq j, \\ -j/\sqrt{j(j+1)}, & r = j+1, \\ 0, & r > j+1. \end{cases}$$

For any strictly positive $\psi_{bi} \in \Delta^{S_b-1}$, the ILR coordinate is $\mathbf{y}_{bi} = Q_b' \log \psi_{bi} \in \mathbb{R}^{d_b}$, where the logarithm is elementwise. For $S_b = 2$, $Q_b = (1, -1)'/\sqrt{2}$ and $y_{bi} = \log(\psi_{bi1}/\psi_{bi2})/\sqrt{2}$. Thus a two-path ILR test is a scaled log-odds, or differential-PSI, test.

Statistical Model

Paired-primer EC likelihood

For block b , pseudobulk i , and primer p , define $N_{bip} = \mathbf{1}_{E_{bp}}' \mathbf{c}_{bip} \in \mathbb{N}_0$, $\mathbf{q}_{bip} = A_{bp} \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{E_{bp}}$, and $Z_{bip} = \mathbf{1}_{E_{bp}}' \mathbf{q}_{bip} \in \mathbb{R}_+$. Conditional on the primer-specific UMI total,

$$\mathbf{c}_{bip} \mid N_{bip}, \boldsymbol{\theta}_{bi} \sim \text{Multinomial}(N_{bip}, \mathbf{q}_{bip}/Z_{bip}).$$

The oligo(dT) and random-hexamer protocols use separate A_{bp} matrices and likelihood terms. Their different isoform exposures therefore model primer-specific positional coverage without forcing a common bias profile.

The genome-wide hierarchical model supplies $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$. Let $s_b(t) \in \{1, \dots, S_b\}$ map isoform $t \in \mathcal{I}_b^{\text{sp}}$ to its path. The baseline spliced-isoform mass is the scalar $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \theta_{bit}^{(0)} \in \mathbb{R}_+$. For $t \in \mathcal{I}_b^{\text{sp}}$, local refinement uses

$$\theta_{bit}(\boldsymbol{\delta}_{bi}) = \omega_{bi}^{\text{sp}} \frac{\theta_{bit}^{(0)} \exp\{(Q_b)_{s_b(t), \cdot} \boldsymbol{\delta}_{bi}\}}{\sum_{u \in \mathcal{I}_b^{\text{sp}}} \theta_{biu}^{(0)} \exp\{(Q_b)_{s_b(u), \cdot} \boldsymbol{\delta}_{bi}\}},$$

where $(Q_b)_{s, \cdot} \in \mathbb{R}^{1 \times d_b}$. Nuisance-isoform abundances are unchanged. This preserves total spliced mass and the relative isoform mixture within each path while estimating only $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$.

Path response and inferential covariance

Each column of $C_b \in \{0, 1\}^{S_b \times T_g}$ corresponding to $t \in \mathcal{I}_b^{\text{sp}}$ contains one 1 in row $s_b(t)$; nuisance isoform columns are zero. Define

$$\mathbf{a}_{bi} = C_b \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{S_b}, \quad \boldsymbol{\psi}_{bi} = \frac{\mathbf{a}_{bi}}{\mathbf{1}_{S_b}' \mathbf{a}_{bi}} \in \Delta^{S_b-1}, \quad \mathbf{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}.$$

The perturbation parameterization implies $\mathbf{y}_{bi} = \mathbf{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$, where $\mathbf{y}_{bi}^{(0)} \in \mathbb{R}^{d_b}$ is computed from the baseline.

Let $I_{\delta,bi} \in \mathbb{S}_+^{d_b}$ be the expected Fisher information of the summed paired-primer log likelihood with respect to $\boldsymbol{\delta}_{bi}$. The conditional covariance is $V_{bi}^{\text{cond}} = I_{\delta,bi}^{-1} \in \mathbb{S}_+^{d_b}$ when the information is positive definite.

The profile calculation instead allows all isoform directions to vary. Set $\boldsymbol{\eta}_{bi} = \log \boldsymbol{\theta}_{bi} \in \mathbb{R}^{T_g}$, $D_{bi} = \text{diag}(\boldsymbol{\theta}_{bi}) \in \mathbb{R}^{T_g \times T_g}$, $\mathbf{d}_{bp} = A'_{bp} \mathbf{1}_{E_{bp}} \in \mathbb{R}^{T_g}$, and $\mathbf{r}_{0,bip} = (\mathbf{d}_{bp} \odot \boldsymbol{\theta}_{bi}) / Z_{bip} \in \mathbb{R}^{T_g}$. The primer-specific expected information is

$$I_{\eta,bip} = N_{bip} \left[\frac{D_{bi} A'_{bp} \text{diag}(\mathbf{q}_{bip}^{-1}) A_{bp} D_{bi}}{Z_{bip}} - \mathbf{r}_{0,bip} \mathbf{r}'_{0,bip} \right] \in \mathbb{S}_+^{T_g}.$$

Let $J_{bi} = \partial \mathbf{y}_{bi} / \partial \boldsymbol{\eta}'_{bi} \in \mathbb{R}^{d_b \times T_g}$. Then

$$V_{bi}^{\text{prof}} = J_{bi} \left(\sum_{p=1}^P I_{\eta,bip} \right)^+ J'_{bi} \in \mathbb{S}_+^{d_b}.$$

If a row direction of J_{bi} overlaps the information null space, that contrast is marked unidentified rather than assigned zero variance by the pseudoinverse.

Cell-resolved likelihood

For cell h in pseudobulk i , let $\mathbf{c}_{bihp} \in \mathbb{N}_0^{E_{bp}}$, $N_{bihp} = \mathbf{1}'_{E_{bp}} \mathbf{c}_{bihp} \in \mathbb{N}_0$, and $\boldsymbol{\theta}_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$ be its EC counts, total, and baseline. Every cell shares the same pseudobulk perturbation $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$, but its isoform vector $\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi}) \in \mathbb{R}_+^{T_g}$ is normalized using its own baseline. The objective is the scalar

$$\ell_{bi}(\boldsymbol{\delta}_{bi}) = \sum_{h=1}^{H_{bi}} \sum_{p=1}^P \log \text{Multinomial} \left(\mathbf{c}_{bihp}; N_{bihp}, \frac{A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})}{\mathbf{1}'_{E_{bp}} A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})} \right).$$

Thus cells contribute assignment information but do not become biological replicates. Let $U_{bih} \in \mathbb{R}_+$ be the cell's total UMI count and $g_{bih} \in [0, 1]$ its decoded proportion for the block's gene. The scalar weight is $w_{bih} = U_{bih} g_{bih}$, and the reported pseudobulk composition is

$$\boldsymbol{\psi}_{bi} = \frac{\sum_{h=1}^{H_{bi}} w_{bih} \boldsymbol{\psi}_{bih}}{\sum_{h=1}^{H_{bi}} w_{bih}} \in \Delta^{S_b-1}.$$

Cell- and primer-specific Fisher information matrices in $\mathbb{S}_+^{d_b}$ are summed, inverted, and propagated through this weighted average to obtain $V_{bi} \in \mathbb{S}_+^{d_b}$.

Effective counts and compositional regression

Treating all gene UMIs as known path assignments is overconfident when ECs are ambiguous. For each fitted $\boldsymbol{\psi}_{bi}$, define the unit-multinomial ILR covariance

$$G_{bi} = Q'_b \text{diag}(\boldsymbol{\psi}_{bi}^{-1}) Q_b \in \mathbb{S}_+^{d_b}.$$

The scalar effective count matches G_{bi}/n to the EC-derived V_{bi}^{cond} in Frobenius inner product $\langle U, W \rangle_F = \text{tr}(U'W)$, for $U, W \in \mathbb{R}^{d_b \times d_b}$:

$$n_{bi}^{\text{eff}} = \min \left\{ N_{bi}^{\text{gene}}, \frac{\langle G_{bi}, G_{bi} \rangle_F}{\langle G_{bi}, V_{bi}^{\text{cond}} \rangle_F} \right\} \in \mathbb{R}_+,$$

where $N_{bi}^{\text{gene}} \in \mathbb{N}_0$ is the gene UMI count. Fractional path counts are $\mathbf{x}_{bi} = n_{bi}^{\text{eff}} \boldsymbol{\psi}_{bi} \in \mathbb{R}_+^{S_b}$, with $\mathbf{1}_{S_b}' \mathbf{x}_{bi} = n_{bi}^{\text{eff}}$.

For one block, stack $n = n_b \in \mathbb{N}$ observations in a design $X \in \mathbb{R}^{n \times k}$, where $k \in \mathbb{N}$, and coefficient matrix $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$. The mean composition $\boldsymbol{\mu}_{bi} \in \Delta^{S_b-1}$ is defined by reference-path logits

$$\log \frac{\mu_{bis}}{\mu_{biS_b}} = (X\mathbf{B}_b)_{is}, \quad s = 1, \dots, d_b.$$

The generalized Dirichlet–multinomial likelihood, extended to fractional counts through gamma functions, is

$$\mathbf{x}_{bi} \mid n_{bi}^{\text{eff}} \sim \text{DirichletMultinomial} \left(n_{bi}^{\text{eff}}, \kappa_b \boldsymbol{\mu}_{bi} \right),$$

where $\kappa_b \in \mathbb{R}_+ \cup \{\infty\}$ is a block-level concentration. Larger κ_b means less overdispersion; the endpoint $\kappa_b = \infty$ is exactly multinomial.

For nested designs $X_0 \in \mathbb{R}^{n \times k_0}$ and $X_1 \in \mathbb{R}^{n \times k_1}$, with $k_1 \geq k_0$ and the first $k_0, k_1 \in \mathbb{N}$, and the first k_0 columns of X_1 equal to X_0 , let $\hat{\ell}_0, \hat{\ell}_1 \in \mathbb{R}$ be maximized scalar log likelihoods. The statistic and nominal degrees of freedom are

$$\Lambda_b = 2(\hat{\ell}_1 - \hat{\ell}_0) \in \mathbb{R}_+, \quad \nu_b = (k_1 - k_0)d_b \in \mathbb{N}_0.$$

The fixed-concentration analysis estimates $\hat{\kappa}_{0b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_0 and uses it under X_1 . The free-concentration sensitivity estimates $\hat{\kappa}_{1b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_1 . In the latter case both models contain one concentration nuisance parameter, so ν_b is unchanged. Both finite and exact multinomial endpoints are evaluated.

Gaussian mouse-replicate regression

For a condition test within one block and cell type, stack $\hat{\mathbf{y}}_{bi} \in \mathbb{R}^{d_b}$ over $n \in \mathbb{N}$ mice. Let $X_b \in \mathbb{R}^{n \times k}$, $k \in \mathbb{N}$, contain an intercept and condition indicators, $\mathbf{B}_b^{\text{GLS}} \in \mathbb{R}^{k \times d_b}$ be regression coefficients, and $\boldsymbol{\epsilon}_{bi} \in \mathbb{R}^{d_b}$ be residuals:

$$\hat{\mathbf{y}}_{bi} = (X_{b,i} \cdot \mathbf{B}_b^{\text{GLS}})' + \boldsymbol{\epsilon}_{bi}, \quad \text{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I_{d_b} \in \mathbb{S}_+^{d_b}.$$

Here $X_{b,i} \in \mathbb{R}^{1 \times k}$, and $\tau_b^2 \in \mathbb{R}_+$ is residual between-mouse variance estimated by REML under the null. If $r_b \in \mathbb{N}$ non-reference condition columns are tested, their coefficients form $\boldsymbol{\beta}_b^{\text{test}} \in \mathbb{R}^{r_b d_b}$ with covariance $\Sigma_b^{\text{test}} \in \mathbb{S}_+^{r_b d_b}$. The Wald statistic is

$$W_b = (\boldsymbol{\beta}_b^{\text{test}})' (\Sigma_b^{\text{test}})^{-1} \boldsymbol{\beta}_b^{\text{test}} \in \mathbb{R}_+, \quad \nu_b^{\text{GLS}} = r_b d_b.$$

Paired cell-type test and permutations

For a cell-type test, let $M_b, L_b \in \mathbb{N}$ be the numbers of represented biological mice and cell types, respectively. The null design $X_{0b} \in \{0, 1\}^{n_b \times M_b}$ contains an intercept-coded set of mouse fixed effects. The alternative $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ appends treatment-coded cell-type indicators. Consequently $k_0 = M_b$, $k_1 = M_b + L_b - 1$, and $\nu_b = (L_b - 1)d_b$. The tested coefficient block is $\mathbf{B}_b^{\text{cell}} \in \mathbb{R}^{(L_b - 1) \times d_b}$, and

$$H_{0b} : \mathbf{B}_b^{\text{cell}} = 0 \quad \text{versus} \quad H_{1b} : \mathbf{B}_b^{\text{cell}} \neq 0.$$

Mouse effects absorb condition and every shift shared by cell types from the same mouse.

For $R = 500 \in \mathbb{N}$ permutations and $r \in \{1, \dots, R\}$, cell-type labels are shuffled only among rows from the same mouse, producing $X_{1b}^{(r)} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ and statistic $\Lambda_b^{(r)} \in \mathbb{R}_+$. Let $\varepsilon_b = 10^{-6} \max(1, |\Lambda_b|) \in \mathbb{R}_+$. The finite-permutation p-value is

$$p_b = \frac{1 + \sum_{r=1}^R \mathbb{I}\{\Lambda_b^{(r)} \geq \Lambda_b - \varepsilon_b\}}{R + 1} \in (0, 1], \quad R = 500.$$

The null coefficients and $\hat{\kappa}_{0b}$ are reused, but each permuted alternative is optimized. The tolerance counts numerically equal likelihood optima as ties. Benjamini–Hochberg correction is applied across canonical partitions with at least 30 pseudobulk rows.

Joint condition tests

ACAT combines $J \in \mathbb{N}$ component p-values $p_j \in (0, 1)$, $j = 1, \dots, J$, into the scalar

$$p_{\text{ACAT}} = \frac{1}{2} - \frac{1}{\pi} \arctan \left[\frac{1}{J} \sum_{j=1}^J \tan\{\pi(1/2 - p_j)\} \right] \in (0, 1).$$

One level combines cell-type-specific p-values for a path partition; a second combines distinct partitions within a gene.

The shared-effect Gaussian model for block b uses $M \in \mathbb{N}$ mice indexed by $m \in \{1, \dots, M\}$, and $L_m \in \mathbb{N}$ cell types for mouse m , indexed by $c \in \{1, \dots, L_m\}$. Its response is $\mathbf{y}_{bmc} \in \mathbb{R}^{d_b}$, design row $X_{bmc} \in \mathbb{R}^{1 \times k}$, where $k \in \mathbb{N}$, coefficient matrix $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$, mouse random effect $\mathbf{u}_{bm} \in \mathbb{R}^{d_b}$, and residual $\mathbf{e}_{bmc} \in \mathbb{R}^{d_b}$:

$$\mathbf{y}_{bmc} = (X_{bmc} \mathbf{B}_b)' + \mathbf{u}_{bm} + \mathbf{e}_{bmc}.$$

The random terms satisfy $\mathbb{E}(\mathbf{u}_{bm}) = \mathbb{E}(\mathbf{e}_{bmc}) = \mathbf{0}_{d_b}$, $\text{Var}(\mathbf{u}_{bm}) = \tau_{\text{mouse}}^2 I_{d_b}$, and $\text{Var}(\mathbf{e}_{bmc}) = V_{bmc} + \tau_{\text{resid}}^2 I_{d_b}$. They are independent across mice, and residuals are independent across cell types conditional on $\mathbf{u}_{bm}$. Hence, for $c \neq c'$,

$$\text{Cov}(\mathbf{y}_{bmc}, \mathbf{y}_{bmc'}) = \tau_{\text{mouse}}^2 I_{d_b}, \quad \text{Var}(\mathbf{y}_{bmc}) = V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2) I_{d_b},$$

where $V_{bmc} \in \mathbb{S}_+^{d_b}$, and $\tau_{\text{mouse}}^2, \tau_{\text{resid}}^2 \in \mathbb{R}_+$ are shared-mouse and residual variance components estimated by two-component REML under the no-condition model. Condition labels are permuted once per mouse and applied to all its cell types.

Algorithms

All algorithms operate independently over canonical blocks unless a joint gene-level combination is stated. “Fit” below means bounded L-BFGS with analytic gradients and a fourfold iteration-limit retry after a warm start from the first endpoint.

Algorithm 1: construct splice blocks and path maps

Input: for one gene, $K_g \in \mathbb{N}$ annotated spliced isoforms, their ordered zero-based half-open exon intervals, and the retained model set $\mathcal{I}_g$ of size T_g . **Output:** blocks b , maps $C_b \in \{0, 1\}^{S_b \times T_g}$, and bases $Q_b \in \mathbb{R}^{S_b \times (S_b - 1)}$.

1. Split the gene at every annotated exon boundary to obtain atomic genomic intervals.

2. Mark intervals exonic in every spliced isoform; merge adjacent marked intervals into constitutive anchors.
3. For each consecutive anchor pair, record each isoform's ordered chain of intervening exonic intervals. Use terminal pseudo-anchors when no shared first or last anchor exists.
4. Assign identical chains to the same path, then restrict the path map to $\mathcal{I}_b^{\text{sp}}$, the annotated isoforms also present in $\mathcal{I}_g$. Remap represented path labels, discard a block if fewer than two paths remain, and embed the one-hot assignments in C_b ; columns for $\mathcal{I}_g \setminus \mathcal{I}_b^{\text{sp}}$ are zero.
5. Construct Q_b from the Helmert formula above.
6. Canonicalize the unordered collection of isoform sets assigned to paths. Retain one representative for identical partitions and attach all equivalent interval identifiers to it.

Algorithm 2: pseudobulk local fit and covariance

Input: $c_{bip} \in \mathbb{N}_0^{E_{bp}}$, $A_{bp} \in \mathbb{R}_+^{E_{bp} \times T_g}$, baseline $\theta_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$, C_b , and Q_b . **Output:** $\hat{\psi}_{bi} \in \Delta^{S_b-1}$, $\hat{y}_{bi} \in \mathbb{R}^{d_b}$, and conditional/profile covariances in $\mathbb{S}_+^{d_b}$.

1. Initialize $\delta_{bi} \in [-20, 20]^{d_b}$ at zero; for $S_b = 2$, also evaluate a fixed log-ratio grid and retain its best point.
2. Form $\theta_{bi}(\delta_{bi})$ while preserving spliced mass and within-path isoform mixtures.
3. Sum the two primer-specific multinomial log likelihoods and optimize over δ_{bi} .
4. Map the fitted isoform vector through C_b , normalize to $\hat{\psi}_{bi}$, and compute $\hat{y}_{bi} = Q_b' \log \hat{\psi}_{bi}$.
5. Sum primer Fisher information for δ_{bi} and invert it to obtain V_{bi}^{cond} . Mark the response unidentified if the information is singular in a tested path direction.
6. For the profile sensitivity, sum $I_{\eta, bip} \in \mathbb{S}_+^{T_g}$, compute $J_{bi} \in \mathbb{R}^{d_b \times T_g}$, and form V_{bi}^{prof} . Apply the information-null-space check.
7. Write every estimate and its convergence, identifiability, and reliability flags; do not silently replace a failed covariance.

Algorithm 3: cell-resolved local fit

Input: cell EC vectors $c_{bihp} \in \mathbb{N}_0^{E_{bp}}$, cell baselines $\theta_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$, weights $w_{bih} \in \mathbb{R}_+$, and the same A_{bp}, C_b, Q_b . **Output:** one pseudobulk response and covariance with dimensions identical to Algorithm 2.

1. Allocate one shared $\delta_{bi} \in \mathbb{R}^{d_b}$; do not allocate a free effect per cell.
2. For every cell and primer, perturb its own baseline, normalize its EC probabilities, and add its multinomial log likelihood.
3. Optimize the summed objective over the shared δ_{bi} .
4. Compute each fitted cell composition $\hat{\psi}_{bih} \in \Delta^{S_b-1}$ and take the normalized w_{bih} -weighted average.

5. Sum the d_b -by- d_b cell-by-primer Fisher matrices, invert, and propagate through the weighted composition average.
6. Send the resulting pseudobulk response to exactly the same mouse-replicate test as Algorithm 4.

Algorithm 4: Gaussian condition and cell-type tests

Input: n responses $\hat{\mathbf{y}}_{bi} \in \mathbb{R}^{d_b}$, covariances $V_{bi} \in \mathbb{S}_+^{d_b}$, and metadata for their mice, conditions, and cell types. **Output:** one Wald p-value for a condition-within-cell-type hypothesis or one paired cell-type hypothesis.

1. For a condition test, retain conditions with at least three biological mice. Construct $X_b \in \mathbb{R}^{n \times k}$ from an intercept and condition indicators, and form its reduced design $X_{0b} \in \mathbb{R}^{n \times (k-r_b)}$ by deleting the $r_b \in \mathbb{N}$ tested condition columns.
2. For a cell-type test, let the $n = n_b$ rows represent M_b mice and L_b cell types. Construct $X_{0b} \in \{0, 1\}^{n_b \times M_b}$ from mouse effects and $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ by appending $L_b - 1$ cell-type indicators. Require the applicable full and reduced designs to have full column rank.
3. Under the reduced model, estimate $\tau_b^2 \in \mathbb{R}_+$ by REML using covariance $V_{bi} + \tau_b^2 I_{d_b}$.
4. Hold $\hat{\tau}_b^2$ fixed and fit the applicable full design by GLS. Its coefficient matrix is $k \times d_b$ for the condition model and $(M_b + L_b - 1) \times d_b$ for the cell-type model.
5. Stack the tested coefficients: $r_b d_b$ entries for condition or $(L_b - 1) d_b$ for cell type. Compute W_b and compare it with the corresponding chi-squared reference distribution.
6. For calibration, permute condition once per mouse for a condition test, or shuffle cell-type labels only within mouse for a cell-type test. Hold $\hat{\tau}_b^2$ fixed and refit each alternative.
7. Apply BH separately by biological question and by conditional versus profile covariance.

Algorithm 5: paired compositional cell-type test

Input: path estimates and conditional covariances from Algorithm 2. **Output:** one exact permutation p-value per canonical block.

1. Keep converged, conditionally identifiable estimates. Require each modeled path proportion to be at least 0.05 and each represented cell type to occur in at least three mice.
2. Convert each $(\psi_{bi}, V_{bi}^{\text{cond}})$ pair to $n_{bi}^{\text{eff}} \in \mathbb{R}_+$ and $\mathbf{x}_{bi} \in \mathbb{R}_+^{S_b}$.
3. Build $X_{0b} \in \mathbb{R}^{n_b \times M_b}$ from mouse fixed effects and $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ by appending cell-type effects.
4. Fit the null coefficients and $\hat{\kappa}_{0b}$. Fit the observed alternative and compute Λ_b .
5. For the fixed- κ approach, set $\kappa_{1b} = \hat{\kappa}_{0b}$. For the free- κ sensitivity, optimize $\hat{\kappa}_{1b}$ and compare the best finite fit with the exact multinomial endpoint.
6. Repeat $R = 500$ times: shuffle cell-type labels within each mouse, rebuild only the appended columns of $X_{1b}^{(r)}$, reuse the null fit, refit the alternative, and store $\Lambda_b^{(r)}$.
7. Compute the tolerance-aware finite-permutation rank p_b . Require $n_b \geq 30$, then apply BH across canonical blocks.

Algorithm 6: pairwise and dominant-path sensitivities

Input: the same path estimates as Algorithm 5. **Output:** calibrated sensitivity tables, not additional primary discoveries.

1. *Pairwise:* for each cell-type pair, retain only mice containing both types. Use $X_0 \in \mathbb{R}^{2M \times M}$ for M mouse effects and $X_1 \in \mathbb{R}^{2M \times (M+1)}$ for one cell-type effect. Permute the two labels within mouse and correct across every block-by-pair hypothesis.
2. *Dominant paths:* rank paths by the scalar aggregate $\sum_i N_{bi}^{\text{gene}} \psi_{bis}$. Select index set $\mathcal{S}_b \subseteq \{1, \dots, S_b\}$, with $|\mathcal{S}_b| = 2$, without using labels.
3. Let $E_b \in \{0, 1\}^{S_b \times 2}$ select these paths and let $Q_b^{(2)} \in \mathbb{R}^{2 \times 1}$ be their Helmert basis. Transform covariance with $\mathcal{R}_b = (Q_b^{(2)})' E_b' Q_b \in \mathbb{R}^{1 \times d_b}$ and $V_{bi}^{(2)} = \mathcal{R}_b V_{bi} \mathcal{R}_b' \in \mathbb{R}^{1 \times 1}$.
4. Renormalize the two retained proportions, then run Algorithm 5. Report path-floor sweeps as sensitivity analyses.

Algorithm 7: joint condition tests

Input: calibrated block-by-cell-type p-values or ILR responses. **Output:** partition- or gene-level joint p-values.

1. *ACAT across cell types:* collect the J valid p-values for one canonical partition and evaluate p_{ACAT} . Optionally apply the same formula to distinct partition p-values within a gene.
2. *Naive shared effect, negative control:* fit cell-type intercepts and one condition coefficient while treating rows as independent. Retain this only to measure pseudoreplication inflation.
3. *Clustered shared GLS:* construct the block covariance whose diagonal blocks are $V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2) I_{d_b}$ and whose same-mouse off-diagonal blocks are $\tau_{\text{mouse}}^2 I_{d_b}$. Estimate the two scalar variances under the null by REML.
4. Fit the shared condition coefficient by GLS. Permute condition once per mouse, synchronously across all cell types, and refit.
5. Apply BH to the resulting canonical-partition family; do not combine discoveries from negative-control and calibrated methods.

Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced isoform are merged into constitutive anchors. Consecutive anchors define a block, and an isoform path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor features remain nuisance features in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated isoforms, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of isoform sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

Pseudobulks and local fitting

Cell isoform proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible isoforms map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $\|\delta_{bi}\|_\infty \leq 20$ to prevent floating-point underflow, where $\|\cdot\|_\infty$ is the maximum absolute component. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local isoform matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \hat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS. Here $\lambda_{\max}(V_{bi}) \in \mathbb{R}_+$ is the largest eigenvalue of the $d_b \times d_b$ covariance, and $\hat{\psi}_{bis} \in [0, 1]$ is path s 's fitted proportion.

Results

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 rows represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same isoform-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation $p < 0.05$, and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact $p = 0.032$ but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.

Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate κ also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing κ provides no net power gain here.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML–GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable isoforms, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with 21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman rank correlation 0.958); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk isoform heterogeneity.

Compositional and joint-test assessment

Collapsing equivalent isoform partitions reduced the 829 conditional block-by-cell-type rows to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect rows as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic $p = 6.4 \times 10^{-5}$ but permutation $p = 0.143$. Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet–multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.

Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.